

USING ARTIFICIAL INTELLIGENCE SYSTEM CLASSIFICATION OF GLASS JEWELRY CULTURE OF OC EO AND SA HUYNH

Ngo Ho Anh Khoi ¹*Bui Hoang Bac ²Vo Khuong Duy ¹

¹ Southern Can Tho University
² Hanoi University of Mining and Geology

*nhakhohi@nctu.edu.vn

SUMMARY— In the field of Oc Eo research, in order to confirm the uniqueness of the Oc Eo cultural area as well as its relationship with other ancient glass areas in the world, the research team has conducted a number of scientific studies related to the classification and identification of ancient glass jewelry, through artifacts in both personal collections and public collections (museums, galleries).... of Vietnam. Realizing the overlap and confusion between Oc Eo culture and Sa Huynh culture glass jewelry, we propose to use a machine learning system to automatically distinguish Oc Eo and Sa Huynh glass jewelry, through SEM gemological analysis parameters. We hope this method, which has only flourished in the archaeological industry for more than five years around the world, can help improve the integration of advanced scientific and technical systems into the complex work of Vietnamese archeology. This Automatic Identification System of Oc Eo-Sa Huynh Glass (RAS-OESHG) is released by us free of charge for experts, archaeologists, and those working in the archaeological industry. The project is a collaboration between the UNESCO Center for Research and Conservation of Vietnamese Antiquities, Hanoi University of Mining and Geology, Institute of Archeology and Southern Can Tho University.

Keywords— ancient glass classification, non-destructive electronic scanning technique, SEM, Scanning Electron Microscopy, machine learning in archeology, evolutionary machine learning.

I. INTRODUCTION

Through the process of trading in ancient times and today, ancient glassware (Dong Son, Sa Huynh, Oc Eo) can be found widely in all regions of Vietnam and around the world. From a professional perspective in the field of private antique collections, this phenomenon of ancient jewelry being misclassified and confused is very common, and extremely complex. In the specific case of glass jewelry, it is very often sold under many different names, or jewelry from different cultures (Dong Son, Sa Huynh, Oc Eo) is often labeled under the wrong name. In this article, the research team tries to go deeper into distinguishing between Oc Eo and Sa Huynh glass jewelry using gemological methods, through an integrated artificial intelligence system. Thereby, providing a useful tool for scientists to automatically distinguish between Oc Eo and Sa Huynh glassware as well as distinguish other ancient glass groups in the region, as well as current glass jewelry (antique imitations, new ones...). This time, the research is presented through five sections. First, I would like to present an overview of the research history of the topic. Next, the study describes in detail the process, machinery, implementation methods and theory of building a database of Oc Eo-Sa Huynh ancient glass. Next, the study briefly describes the machine learning experimental process and evaluates the results achieved. Finally, there is the general summary of the article.

II. HISTORY OF RESEARCH TOPIC

The direction of researching ancient glass through gemological methods began around the 1990s, starting from a series of studies by J. Henderson (Henderson, 1990), on ancient Roman glass in England. The distinction of ancient glass lines according to the gemological method is based on characteristics related to the composition and content of elements in the composition. Researchers mostly agree on two basic types of ancient glass, based on the composition created by the glass cooking process. It is a group of glasses of the lead family (*lead-glass*), and the alkaline glass group (*natron-glass*). Lead family of glass, invented by the Mesopotamians, via the Silk Road¹ which was introduced into China, becoming ancient Han glass, also called ancient Oriental glass or ancient Asian glass, although this is not accurate. The alkaline glass group was invented by the Romans and appeared in most other cultures neighboring the Mediterranean such as Egypt, Africa, and Arabia....often called the ancient Roman glass group, also known as ancient Western glass (Tait, 2004). Studies of this type in Asia are very rare, only recently developed in a few studies in Thailand, and China, especially in Vietnam, there have not been any articles on this topic, except for a few individual studies by the French. Much of this research has been conducted in Europe, focusing on alkaline glass groups, and the research system has been relatively mature for more than 30 years. The earliest and most notable is the classification by researcher Sayre (Sayre, 1961), dividing the alkali glass group into two main groups, mainly based on pure composition. First is the lmlp group (*low-magnesia, low-potash glasses*) characterized by low levels of magnesium (Mg) and potassium (K), this group appeared in almost all ancient cultures, including Rome, and was very popular from the 1st millennium BC to the 9th century AD. The second is the hmhp group (*high-magnesia, high-potash glasses*) with high content of magnesium (Mg) and potassium (K), appeared in Central Asia, before the 7th century BC, and appeared in the Arab culture area, before the 9th century BC.

¹ Tait, Hugh, (2004). Five Thousand Years of Glass. University of Pennsylvania Press (orig. British Museum Press). ISBN 978-0-8122-1888-6

A little later, is the classification by researcher Freestones (Freestones, 2005), based on the origin of the composition of Calcium (Ca) to divide into two groups. Scholar Rehren (Rehren, 2015) offers a way to divide the mixture between Freestones and Weepohl into six groups of ancient glass types, divided based on raw material source, including both alkali and lead groups. Many similar studies are based on component division, not very different from the studies mentioned above. Research on the gemological properties of glass jewelry in Vietnam currently includes research mainly by Ngo Ho Anh Khoi (Ngo Ho Anh Khoi, 2019; 2019b). The classification of ancient glass into groups may be a simple method, but today it does not meet modern requirements: each group of ancient glass itself can be divided into many smaller groups of glass, each of which is not completely identical in nature but has differences from each other, although they share the common characteristics of the group as a whole. This distinction is especially problematic when different types of glass in different regions were created in the same tradition, leading to great archaeological differences but from a gemological perspective do not show differences large enough to be detected with rudimentary mathematical tools.

Recently, we have noticed the trend of applying artificial intelligence technologies to the classification, appraisal, evaluation and judgment of antiques, sites and heritage such as Bickler's studies (Bickler, 2018) in the appraisal of ancient porcelain, Jones's research (Jones, 2019) in the classification of ancient plants, Ngo Ho's studies (Ngo Ho, 2013; 2014; 2015; 2017) in classifying and evaluating ancient documents, research is gradually establishing the first automatic identification systems for Vietnamese antiquities through suggesting a system of advanced technologies. In recent research by Ngo Ho (2019; 2020), this study uses the MLP neural network artificial intelligence system to classify Oc Eo ancient glass. After obtaining the feature sets extracted from the SEM system (will be explained in more detail later), including 36 features, the application has built recognition training neural networks, compared with each other to choose the best network. In this application, a network has been selected with: 36 input values, 120 hidden layer neurons, 1 output layer neuron. During the training process, a set of networks suitable for identification was obtained. During the identification process, the application also continues to receive data directly entered by the user, calculate its characteristics and put it into the network for identification. The properties of the analyzed rays are selected with some characteristic properties and vectorized to transform into a 36-dimensional vector $\{x_1, x_2, \dots, x_{36}\}$. The number of networks we install is a system that builds a single network to identify specimens, but can expand many different types of specimens at the same time. The neuron set consists of 3 layers: layer of input values, layer of hidden neurons, layer of output neurons. There are 2 sets of weights at the links from the input layer to the hidden layer and from the hidden layer to the output layer. We, based on the characteristics of artifact identification, have chosen a network architecture type of feedforward neural network combined with error backpropagation algorithm. Thanks to that, results and performance are gradually enhanced during the continuous learning process, improving the accuracy of the final results.

In this paper, the main goal of the research is to improve a recently published system by further integrating different learning methods through the use of evolutionary methods (*evolving learning* hay *continuous learning*). This idea comes from the unique characteristics of stream data in the field of archeology. Unlike traditional datasets, archaeological data accumulates over time as new discoveries arise from excavations. This research poses significant challenges due to data scarcity, which can be attributed to three main reasons. First, real-life situations in archaeological investigations make finding artifacts difficult, and performing experimental excavations to extract characteristics from these artifacts can be very expensive. As a result, data is collected unevenly over time and in limited quantities. Second, although archaeological research institutes allow the use of the results of feature extraction experiments and final models, they do not allow the storage of actual archaeological data unless the classification method requires it. And finally, and most importantly, in the field of archeology, there are cases where artifacts initially classified into a particular group may be reclassified in the future based on new data and new findings. The classification of artifacts into specific categories or classifications depends on many multidisciplinary factors, many of which are relevant to future excavations. Some of these factors may be completely unexpected. This also poses a challenge to the fields of data science and machine learning, known as "concept slippage" (*concept drift*), will be explained in more detail in the next section.

Therefore, with the limitations of traditional learning methods, evolutionary learning methods become even more suitable and attractive to solve this problem. In this study, the research team will implement and compare the results of evolutionary learning methods using the research team's archaeological dataset. A comprehensive overview of evolutionary machine learning methods (*incremental learning*, *evolving learning* hay *continuous learning*) has been fully researched in (Ngo Ho, 2015). Considering the specific context of archaeology, the team realized that the system could effectively use lightweight and simple methods based on a data selection approach that would be easy to set up and convenient in reusing the scant data from excavations. This method ensures that important information is not lost when the generalized concept is continuously applied to evolve by controlling the content of data contained in each learning data package, as analyzed in the

study (Ngo Ho, 2015). To achieve this, the study proposes to use an approach based on data selection in each learning data package called "sliding window" (*sliding windows*) is simple, similar to the FLORA method (Widmer, 1996). This principle involves updating the model at each time t' any using the training data closest to the point in time t' , defined by a sliding window of predetermined size (custom based on time scale or amount of data). This method may include batch learning using selected data in a sliding window or continuously updating the model if possible.

Online learning methods allow. Typically, the methods of this approach include three steps (Bifet, 2007):

1. Detect conceptual change using statistical tests across different windows.
2. If variation is observed, select representative and recent examples to calibrate the models.
3. Update models. The size of the window is predetermined by the user, and each window overlaps the previous window by sharing a batch of data. At each step, a new model is relearned, representing an updated set of classes.

The important aspect of this approach lies in determining the appropriate window size. While most methods use fixed windows configured for each practical problem, there are also methods that aim to automatically detect the optimal analysis window size. For example, in (Bifet, 2007) with ADWIN (ADaptive WINdow), the author tests a set of window sizes by dividing each window into subwindows of minimum size. If the subwindows representing different distributions are large enough, a statistically significant size is considered a good choice. In (Lazarescu, 2003), the author suggests using two models at each step, each trained with a different window size: S (predetermined standard size) and $2S$ (Small window in size S used to detect new concept spaces using statistical testing, while the window is large with size $2S$ used to update the model when new concept space is detected). In (Last, 2002) with OLIN (On Line Information Network), the author proposes to dynamically adjust the window size based on the performance achieved on the validation data set. The new data is divided into two parts: one for training and one for validation. Multiple windows with different sizes are applied independently for learning and testing, and the size that produces the best results on the validation data is selected for the current step. However, to implement this method effectively, it requires performing learning stages on sufficiently large batches of data. Finally, in (Klinkenberg, 2004), the authors apply a successive increase in window sizes. At each step, the performance (in terms of error rate) is calculated for different window sizes, and the size that produces the best performance is chosen (for example, size number 1 represents the last batch, size number 2 represents the last two batches, size number 3 represents the last three batches, and so on). In this article, research will apply Klinkenberg's ideas to classical methods to transform them into evolutionary forms (*evolving learning*), used on archaeological data sets, compared to discover the most optimal method for this problem. The basic reason for choosing Klinkenberg's idea is clear: research can control information loss by managing the "density" of the learning data within the chosen optimal size. By using Klinkenberg's methods, with just 1 simple parameter n is the size of the data batches, while it is possible to control the loss of valuable information caused by machine learning methods attempting to generalize or detail concepts often accompanied by deviations from the true nature of the concept.

III. BUILDING A DATABASE OF VIETNAM ANCIENT GLASS

The Oc Eo and Sa Huynh specimens used in building the database of Vietnamese ancient glass have been exhibited at many national exhibitions and have been appraised by the National Appraisal Council. These specimens belong to the collection of Mr. Nguyen Trong Co and Mr. Ngo Ho Anh Khoi, belonging to the Sa Huynh and Oc Eo cultures. SEM-EDS technique has been presented in detail in many previous research articles (Ngo Ho Anh Khoi, 2019; 2019b), the research would like to be basically reiterated as follows:

A. Backscattered electrons Analysis Method

Scanning electron microscope (SEM) combined with EDS (Model: Quanta 450; Manufacturer: FEI-USA). Electron microscopes can produce high-resolution images of a sample's surface by using a narrow electron beam (beam of electrons) that scans across the sample's surface. Imaging of the specimen is performed through recording and analyzing the radiation emitted from the interaction of the electron beam with the specimen surface. Backscattered electrons are the initial electron beam that bounces back when interacting with the sample surface, so they often have high energy. This scattering depends greatly on the chemical composition of the sample surface, so backscattered electron images are very useful for chemical composition contrast (EDS) analysis. In addition, backscattered electrons can be used to record backscattered electron diffraction images, helping to analyze crystal structure (electron polarization mode).

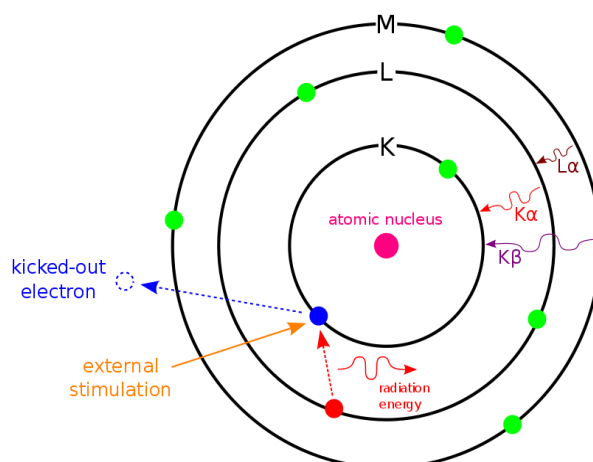


Figure 1. The technique is described in terms of electron interactions

B. Energy Dispersive X-ray Spectroscopy (EDXS)

Energy Dispersive X-ray Spectroscopy (EDXS) or Wavelength Dispersive X-ray Spectroscopy (WDXS). is a technique for analyzing the chemical composition of solid objects based on recording the X-ray spectrum emitted from the solid object due to interaction with radiation (mainly high-energy electron beams in electron microscopes).

Samples are analyzed in a vacuum sample chamber environment. Depending on the type of specimen, use high or low pressure vacuum (e.g. low pressure vacuum mode for biological specimens). Examine high-resolution SEM images of the specimen surface using an electron beam (beam of electrons) emitted from an electron gun (can be thermal emission, or field emission...), then accelerated. Electrons are emitted, accelerated and focused into a narrow electron beam (several hundred Angstrom to several nanometers) thanks to a magnetic lens system, then scanned on the sample surface using electrostatic scanning coils. The resolution (Magnification) of SEM is determined from the size of the focused electron beam. In addition, the resolution of SEM also depends on the interaction between the material at the specimen surface and the electrons.



Figure 2. Experimental machine used by specimens in SEM-EDS gemology

C. Process Description and Analysis Parameters

The sample for SEM analysis is a solid sample, the sample size must match the sample chamber (small sample, $< 3 \text{ cm}^3$). Depending on the characteristics of the sample, the sample will be processed to obtain the best SEM image quality. For non-conductive samples, the sample must be surface coated with Carbon, Platinum or Gold. Avoid surface charge phenomenon that causes glare on the sample surface. The sample is glued on the alloy sample base with specialized carbon adhesive tape and then inserted into the sample chamber.

The location and number of analytical processing times play a very important role in the assessment, and because each analysis of a location is very expensive, it is necessary to analyze the locations that need to be analyzed to minimize data redundancy. Therefore, in this study, to the extent permitted, radioactivity was performed under the following conditions: depending on sample quality, each sample was analyzed 5 times in different positions on the specimen. By selecting heterogeneous positions to conduct analysis to achieve maximum differences. Homogeneous samples may be processed less, or bulk samples may be processed less.

Semi-quantitative elemental composition: The electron beam scanned on the sample surface impacts the electron clouds of the element (electron layers M, L, K...). The reflection component includes the X-ray energy dispersion spectrum captured by the detector to determine the elemental composition. Effective reflection intensity (Net Int) represents the degree of reflection of the electron beam,

Kratio value is the reflected electron density, R is the resolution in microns. After determining the reflected electron density. The elemental composition is calibrated by comparison between the spectrum of the standard sample and the spectrum of the measurement sample. The values Z, R, A, F are calibration values with: Z: is the atomic calibration when comparing between standard sample and sample. A: absorption correction (compensation correction for X-rays released but remaining in the sample). F: fluorescence calibration (calibrates the amount of x-rays released with a high-energy x-ray source).

Table 1.SEM characteristic table

Element 2	Weight3 %	Atomic 4 %	Net Int. 5	Error %6	Kratio7	Z	R	A	F
OK (Class K Oxygen)									
FeK (K-class Iron)									

IV. EVALUATION OF EXPERIMENTAL RESULTS

In this article, the research will experiment on nine different classical machine learning methods with the combination of Klinkenberg's ideas to create evolutionary learning methods (*evolving learning*) measures the effectiveness of the optimal window size, using an archaeological dataset. Experiments will install the evolutionary method using a "sliding window" (*sliding windows*) along with determining the optimal window size based on Klinkenberg's ideas to a total of 9 different machine learning algorithms for experimental comparison, incl. AdaBoost algorithm (Freund, 1995): An ensemble estimator that tunes the classifier on the original data set and further tunes the replicas to focus on difficult cases. Gaussian Naïve Bayes algorithm (Chan, 1979): A variation of Naïve Bayes that follows the Gaussian normal distribution and supports continuous data. Decision Tree Algorithm (Breiman, 1984): A flow tree structure that uses decision rules to divide data based on attribute values. K-Neighbors algorithm (Goldberger, 2005): A method that finds a predetermined number of nearest learning samples in terms of distance to a new point and predicts labels based on these neighbors. Extra Tree Algorithm (Geurts, 2006): An estimator based on making random decisions on subsets of the data set to improve prediction accuracy. Bagging algorithm (Ho, 1998): An ensemble meta-estimator that bases classifiers on random subsets of the data set and combines their predictions. MLP algorithm (Hinton, 1989): Based on neural networks to perform classification and optimize the log loss function. Random Forest Algorithm (Breiman, 2001): A meta-estimator that outputs multiple decision tree classifiers on different subsets of the data set and uses averaging to improve prediction accuracy. Bernoulli Naive Bayes algorithm (Lewis, 1998): A variant of Naive Bayes that works well with transformations for limited binary data. These machine learning methods are combined with "sliding window" methods (*sliding windows*) and integrate Klinkenberg's ideas to transform these machine learning methods into evolutionary machine learning methods (*evolving learning*) to be suitable for specific problems of archaeological data.

The algorithms used in this experiment are implemented using version 0.24.2 of the scikit-learn library, which is implemented with the "sliding window" technique (*sliding windows*) in python. Data is processed in streams using a "sliding window" (*sliding windows*) small review the most recent models with size n The team started with $n = 1$, corresponds to classic online learning, then gradually increases $n > 1$, represents batch learning (*batch learning*). For each method, only the best results of n will be selected for comparison with other machine learning methods. The archaeological data set includes a total of 108 samples, of which 25 belong to the Oc Eo samples and the remaining samples belong to Sa Huynh. Each sample is characterized by 60 features (*features*), represents 6 chemical elements with 10 indices for each element. The dataset is divided into training dataset (70%) and test dataset (30%) with random cross-validation (*cross-validation*). The positions of the data in the training and testing datasets were randomly assigned to each experiment, which was repeated 10 times (*10 times separation train/test set randomly*). Additionally, the training data set was randomly shuffled 10 times to generate different scenarios of lineage learning, demonstrating the model's adaptability to evolutionary data. This means that each learning method with each parameter variation will go through 100 random experiments before obtaining the final comparison data. Results will be averaged to ensure reliability and generalizability of the experiments. All results are achieved based on balanced accuracy (*Balanced Accuracy*), which is a suitable metric to evaluate the performance of binary classifiers, especially in cases where there is imbalance between classes. Balanced accuracy considers the true negative rate (*True negative rate*) and true positive (*True positive rate*), overcoming the limitation of precision calculations often in imbalanced data sets (Ngo Ho, 2013). By looking at the TNR ratios (*True negative rate*) and TPR (*True positive rate*), the balanced accuracy formula provides a more accurate and optimal ratio especially for datasets with imbalance

- 2 Element names are based on electron subclasses
- 3 Mass (%) in the specimen
- 4 Atoms (%) in the sample
- 5 The effective intensity is obtained in the specimen
- 6 Error number
- 7 Density of reflected electrons in the sample

The problem is so large that it is impossible or limited to use measures to create additional simulated data as in the case of archeology. Balanced precision formula $BA(Balance Accuracy)$ as follows: $BA = (TPR + TNR) / 2$.

From the perspective of the average results of BA balancing accuracy, all results achieved high efficiency (mostly over 90%), however, the order of superiority was recorded as follows: Bernoulli NB Classifier (97.03%), Gaussian NB Classifier (96.38%), Extra Tree Classifier (96.36%), Random Forest Classifier (96.19%), Bagging Classifier (96.18%), Adaboost Classifier (95.13%), k-neighbors Classifier (94.64%), Decision Tree Classifier (94.62%), MLP Classifier (88.44%), see figure 3. It is easy to see that the Naives Bayes learning algorithm group (Bernouilli and Gaussian NB) achieved the highest results, followed by the Ensemble learning algorithm group (Extra Tree Classifier, Random Forest Classifier, Bagging Classifier, Adaboost Classifier), and finally the group of other solutions. Other algorithms (k-neighbors Classifier, Decision Tree Classifier, MLP Classifier). The Naives Bayes learning algorithm group (Bernouilli and Gaussian NB) also has a low standard deviation, while the other two groups have higher deviations, especially MLP Classifier up to more than 5%. MLP Classifier has relatively lower results in accordance with theory when this method is sensitive to little data, because accurate adjustment only occurs when a lot of data are in the same alignment, and in conditions of little data, it is easy to happen that these adjustments are less accurate, leading to poor accuracy in each class, resulting in this method also having large variations in bias. Ensemble learning algorithms often have good results, this problem is no exception, the difference in efficiency between this group of algorithms and the group of Naives Bayes learning algorithms (Bernouilli and Gaussian NB) is not really large (mostly < 1%). The effectiveness of the Naives Bayes learning algorithm group (Bernouilli and Gaussian NB) which has a probability-based theory shows that the data isolation between Sa Huynh and Oc Eo ancient glass is clear and the correlation with probability is large, or in other words, these two data classes have a relatively clear isolation in the data space; Otherwise, algorithms based on data subgroup isolation such as Ensemble learning algorithms will probably have better results than Naives Bayes learning algorithms.

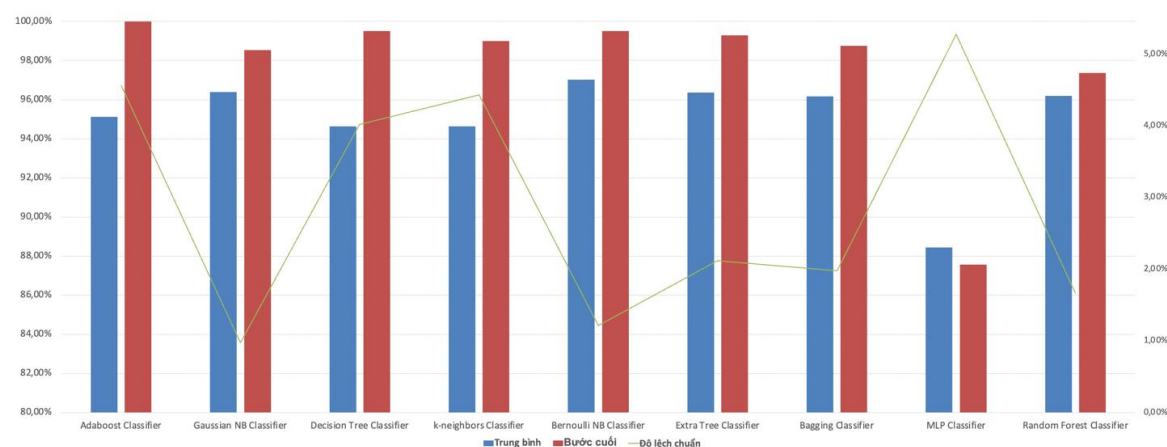


Figure 3.The chart shows the correlation of results between 9 algorithms

This result also shows that, although the data is sparse, these data are relatively evenly represented in the class's concept space (*concept space*), because if there is inequality, the standard deviation in the Ensemble learning algorithm group will certainly be much larger than the standard deviation in the Naives Bayes learning algorithm group. Thanks to such a view, the high effectiveness of k-neighbors Classifier and Decision Tree Classifier, which are based on the division of concept space, have equally good results, which can also be easily explained by the clear isolation of the concept space of both classes. See table 2.

However, there are more or less differences in the average balancing accuracy and the final balancing accuracy. At the final step, the order changes slightly: most methods achieve the same final step, with no really large differences (most are larger than 98%), except for the relatively poor result of MLP Classifier (~ 87%). This shows that the overall results of the algorithms achieved good results thanks to the relatively easy to isolate data set. The only problem with the dataset is that its small number (which cannot be overcome) affects the results of MLP Classifier.

Table 2.BA (Balanced Accuracy) results table shows average, final step and standard deviation.

ALGORITHM	BA Average	THREE Final steps	Standard deviation
Bernoulli NB Classifier	97.03%	99.50%	1,21%
Gaussian NB Classifier	96.38%	98,54%	0,97%

Extra Tree Classifier	96.36%	99,29%	2,12%
Random Forest Classifier	96,19%	97,37%	1,67%
Bagging Classifier	96,18%	98,75%	1,98%

Adaboost Classifier	95,13%	100,00%	4,56%
k-neighbors Classifier	94,64%	99,00%	4,43%
Decision Tree Classifier	94,62%	99,50%	4,02%
MLP Classifier	88,44%	87,56%	5,28%

From the perspective of monitoring the progress of the learning results of the algorithms (see Figure 4). It is easy to see the instability of the MLP Classifier with a relatively large amplitude. The remaining algorithms all have good results in the final steps and there is no big difference. The main difference between the algorithms (except MLP Classifier) is in the early stage (before step 20), when the algorithms are trying to find the best model through data accumulation and selection. Most algorithms gradually stabilize at steps 20-25, and are highly stable at steps 25-30. The large difference in average BA balance accuracy results is mainly affected by the initial unstable period, when the algorithms search for the best model. At this stage, it is easy to see the division into 2 small groups, one group quickly stabilizes early (the Ensemble learning algorithm group and the Naives Bayes learning algorithm group) when reaching stability right in step 5 and gradually increasing efficiency; a group that stabilizes slowly (Decision Tree Classifier, k-neighbors Classifier and MLP Classifier) and increases efficiency relatively slowly and only reaches stability at steps 15-20. That is why there is a disturbance between the average BA balance accuracy results and the final step BA balance accuracy results.

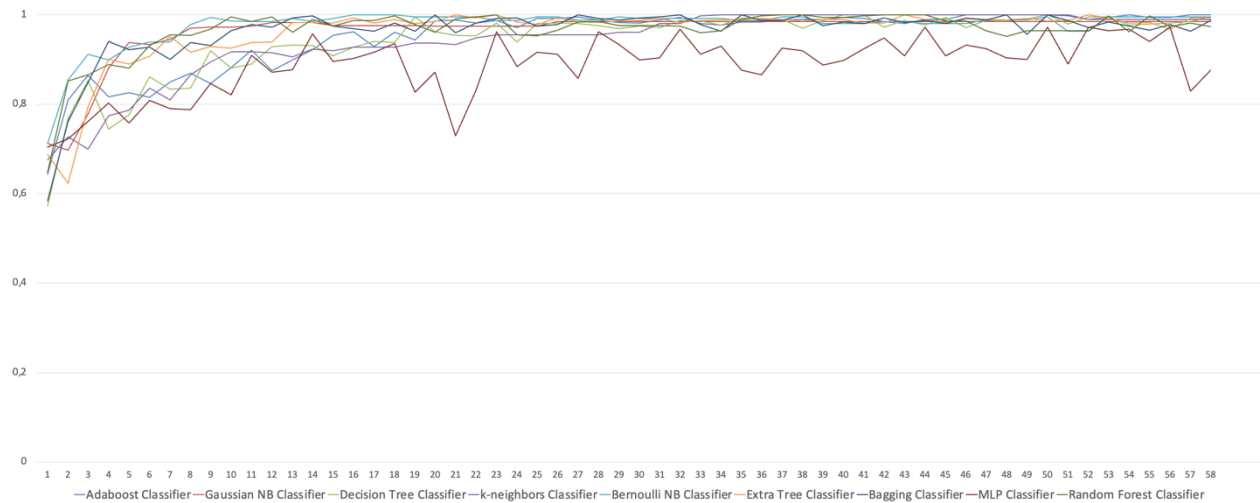


Figure 4.The chart shows the progress of learning results of 9 algorithms

The final result of this problem, Bernoulli NB Classifier gives the best results, both in terms of high balancing accuracy and small standard deviation. This algorithm was installed by the research team into the official system for practical use in this problem. However, the research team also believes that the Extra Tree Classifier algorithm has desirable results, especially in the case of relatively large amounts of data arriving in the future.

V. CONCLUSION

The study was conducted in the unique context of the field of archaeology, where data scarcity and intermittent data flow create a significant challenge for the construction of classification systems. This data shortage stems from three main factors: the difficulty of extracting data on artifacts in real-life contexts, the high costs involved in conducting excavations and extracting characteristics from these objects, and the irregular nature of data collection over time. In the field of archaeology, artifacts initially classified in a particular group belonging to a culture may undergo reclassification based on additional data and new discoveries. Classifying objects or groups into specific categories based on multidisciplinary factors, this complexity makes classification research difficult, as previously conducted research can become outdated and lose its practicality over time. These situations therefore pose challenges for data science fields in archaeology, especially regarding the phenomenon of "concept drift" in data science, which is an important factor driving the application of evolutionary learning methods to archaeological data sets.

The article introduces an artificial intelligence system that allows automatic identification of Oc Eo glass jewelry, through SEM gemological analysis parameters. The system is an attempt to apply high technology to assessment and

artifact identification, a technological step forward compared to other traditional methods, to keep up with the recent remarkable developments in archaeological centers around the world. Artificial intelligence technology integrated into archaeological activities is an inevitable trend of the archaeological industry in the future. The results obtained from the research have been integrated into a system called the System for Automatic Identification of Oc Eo Sa Huynh Glass Jewelry (RAS-OCSHG), completely free for archaeological experts on the system's website. Although this system has good performance in recognition, more support is needed to expand and improve the feature dataset.

VI. REFERENCES

- [1] Anh Khoi Ngo Ho, Hoang Bac Bui, Giang Nguyen Khac, The Luc Trinh, Ancient glass stone in history, Journal of Developmental Philosophy, 2020, accepted.
- [2] Anh Khoi Ngo Ho, Méthodes de classifications dynamiques et incrémentales : application à la numérisation cognitive d'images de documents, Doctoral dissertation, Ecole doctorale Mathématiques, Informatique, Physique Théorique et Ingénierie des Systèmes (Centre-Val de Loire), Tours, 2015.
- [3] Anh Khoi Ngo Ho, Nicolas Ragot, Jean-Yves Ramel, Veronique Eglin, Nicolas Sidere, Document classification in a non-stationary environment: a one class SVM approach, ICDAR 2013, 12th International Conference on Document Analysis and Recognition, Washington DC, USA, 2013.
- [4] Anh Khoi Ngo Ho, Nicolas Ragot, Jean-Yves Ramel, Veronique Eglin, A Multi One-Class Dynamic Classifier for Adaptive Digitization of Documents Streams, International Journal on Document Analysis and Recognition, 2017.
- [5] Anh Khoi Ngo Ho, Trong Co Nguyen, Ngoc Khoi Doan, Hoang Bac Bui, and The Luc Trinh, Distinct gemological properties of Sa Huynh glass ornaments, International Workshop On Heritage Values of Ly Son - Sa Huynh Geopark, Quang Ngai, 2019.
- [6] Anh-Khoi Ngo-Ho, Van-Phuc Pham, Van-Trieu Pham, Dinh-Phung Le, Trong-Co Nguyen, Ngoc-Khoi Doan, Thi-Thuy Tran, Artificial intelligence system for automatic identification of antique Oc Eo Glass jewelry, National Conference on Oc Eo Runes: Ba The, Nen Chua - Excavation, research, conservation and promotion of values, An Giang, Vietnam, 2019b.
- [7] Bickler, Simon H. "Machine Learning Identification and Classification of Historic Ceramics." *Archaeology in New Zealand* 61.1 (2018): 21–33. Print.
- [8] D. Rosenow, Th Rehren Herding cats – Roman to Late Antique glass groups from Bubastis, northern Egypt J. Archaeol. Sci., 49 (2014), pp. 170-184.
- [9] I. Freestone Glass production in Late Antiquity and the Early Islamic period: a geochemical perspective M. Maggetti, B. Messiga (Eds.), *Geomaterials in Cultural Heritage*, Geological Society London Spec Publ 257 (2006), pp. 201-216
- [10] I. Freestone, Leslie, Thirlwall and Goren-Rosen 2003; Freestone, Wolf and Thirlwall 2005.
- [11] I. Freestone, R. Jackson-Tal, O. Tal Raw glass and the production of glass vessels at Late Byzantine Apollonia-Arsuf, Israel J. Glass Stud., 50 (2008), pp. 67-80
- [12] I. C. Freestone, R. Greenwood, Y. Gorin-Rosen Byzantine and early Islamic glassmaking in the Eastern Mediterranean: production and distribution of primary glass G. Kordas (Ed.), *Hyalos – Vitrum – Glass. History Technology and Conservation of Glass and Vitreous Materials in the Hellenic World. 1st International Conference Rhodes – Greece 1–4 April 2001, Athens (2002)*, pp. 167-174
- [13] Jones, B., & Bickler, S. H. (2019). Multi-Scalar and Semi-Automatic Approaches to Detect Archaeological Features in NZ Using Airborne LiDAR Data. *Archaeology in New Zealand*.
- [14] M.-D. Nenna Ateliers de production et sites de consommation en Égypte, Ve s. av. J.-C. Ville s. apr. J.-C Annales du 14e congrès de l'AIHV: Venise-Milan 1998 (2000), pp. 20-24
- [15] N. Kato, I. Nakai, Y. Shindo Change in chemical composition of early Islamic glass excavated in Raya, Sinai Peninsula, Egypt: on-site analysis using a portable X-ray fluorescence spectrometer J. Archaeol. Sci., 36 (2009), pp. 1698-1707
- [16] N. Kato, I. Nakai, Y. Shindo Transitions in Islamic plant-ash glass vessels: on-site chemical analyses conducted at the Raya/al-Tur area on the Sinai Peninsula in Egypt J. Archaeol. Sci., 37 (2010), pp. 1381-1395
- [17] R. Abd-Allah Chemical characterisation and manufacturing technology of late Roman to early Byzantine

glass from Beit Ras/Capitolias, Northern Jordan J. Archaeol. Sci., 37 (2010), pp. 1866-1874

[18] Sayre E. V. and Smith R. W., 1961. Compositional categories of ancient glass. Science 133, 1824–6.

[19] Tait, Hugh, (2004). Five Thousand Years of Glass. University of Pennsylvania Press (orig. British Museum Press). ISBN 978-0-8122-1888-6.

[20] Tait, Hugh, (2004). Five Thousand Years of Glass. University of Pennsylvania Press (orig. British Museum Press). ISBN 978-0-8122-1888-6

[21] Th. Rehren, A. Cholakova The Early Byzantine HIMT Glass from Dichin, Northern Bulgaria Interdiscip. Stud., 22/23 (2010), pp. 81-96

[22] Th. Rehren, Ian C. Freestone, Ancient glass: from kaleidoscope to crystal ball, Journal of Archaeological Science.

Volume 56, April 2015, Pages 233-241

[23] Trương Quốc Bảo, Đào Thị Xuyên, Võ Văn Phúc, New features set for signature recognition using neural network, Vietnam Conference on Mechatronics (VCM), 2014.

- [24] Wedepohl K. H. and Baumann, A., 2000. The use of marine molluscan shells for Roman glass and local raw glass production in the Eifel area (Western Germany). *Naturwissenschaften* 87, 129–32.
- [25] G. Widmer, M. Kubat ; Learning In The Presence Of Concept Drift And Hidden Contexts ; *Machinelearning*, Vol. 23, No. 1, Pp. 69-101, (1996).
- [26] A. Bifet, R. Gavaldà ; Learning From Time-Changing Data With Adaptive Windowing ; In: *Sdm*, Pp. 443–448 (2007).
- [27] M. Lazarescu, S. Venkatesh, H. Bui ; Using Multiple Windows To Track Concept Drift ; *Ida8(1)*, Pp.29–59 (2003).
- [28] M. Last ; Online Classification Of Nonstationary Data Streams ; *Intell. Data Anal.* 6(2) 129–147(2002).
- [29] R. Klinkenberg ; Learning Drifting Concepts: Example Selection Vs. Example Weighting ; *Intelligent Data Analysis, Special Issue On Incremental Learning Systems Capable Of Dealing With Concept Drift*, 8 (3), (2004).
- [30] Y. Freund, R. Schapire, “A Decision-Theoretic Generalization of on-Line Learning and an Application to Boosting”, 1995.
- [31] Zhu, H. Zou, S. Rosset, T. Hastie, “Multi-class AdaBoost”, 2009.
- [32] Chan, Golub, and LeVeque, Updating formulae and a pairwise algorithm for computing sample variances, *Stanford CS tech report STAN-CS-79-773*, Stanford University, 1979.
- [33] L. Breiman, J. Friedman, R. Olshen, and C. Stone, “Classification and Regression Trees”, Wadsworth, Belmont, CA, 1984.
- [34] J. Goldberger, S. Roweis, G. Hinton, R. Salakhutdinov, “Neighbourhood Components Analysis”, *Advances in Neural Information Processing Systems*, Vol. 17, May 2005, pp. 513-520.
- [35] P. Geurts, D. Ernst., and L. Wehenkel, “Extremely randomized trees”, *Machine Learning*, 63(1), 3-42, 2006.
- [36] L. Breiman, “Pasting small votes for classification in large databases and on-line”, *Machine Learning*, 36(1), 85- 103, 1999.
- [37] L. Breiman, “Bagging predictors”, *Machine Learning*, 24(2), 123-140, 1996.
- [38] T. Ho, “The random subspace method for constructing decision forests”, *Pattern Analysis and Machine Intelligence*, 20(8), 832-844, 1998.
- [39] G. Louppe and P. Geurts, “Ensembles on Random Patches”, *Machine Learning and Knowledge Discovery in Databases*, 346-361, 2012.
- [40] Hinton, Geoffrey E. “Connectionist learning procedures.” *Artificial intelligence* 40.1 (1989): 185-234.
- [41] L. Breiman, “Random Forests”, *Machine Learning*, 45(1), 5-32, 2001.
- [42] David D. Lewis, Naïve (Bayes) at forty: The independence assumption in information retrieval, *Machine Learning: ECML-98, Lecture Notes in Computer Science book series (LNAI, volume 1398)*, 1998.

USING ARTIFICIAL INTELLIGENCE TO CLASSIFY OC EO AND SA HUYNH GLASS JEWELRY

Ngo Ho Anh Khoi ¹*Bui Hoang Bac ²Vo Khuong Duy ¹

¹ Nam Can Tho University
² Hanoi University of Mining and Geology

ABSTRACT— In the field of Oc Eo research, in order to confirm the uniqueness of the Oc Eo cultural region as well as its relationship with other ancient glass regions around the world, we have conducted a number of scientific studies related to the classification and identification of ancient glass jewelry, through artifacts found in both private collections and public collections (museums, galleries, etc.) in Vietnam. Recognizing the overlap and confusion between Oc Eo and Sa Huynh glass jewelry, we propose using a machine learning system to automatically distinguish between Oc Eo and Sa Huynh glass jewelry, based on SEM gemological analysis parameters. We hope that this method, which has only recently flourished in the field of archaeology over the past five years, can help to improve the integration of advanced scientific and technological systems into the complex tasks of Vietnamese archaeology. This Oc Eo-Sa Huynh Ancient Glass Automatic Recognition System (RAS-OESHG) is released free of charge to experts, archaeologists, and those working in the field of archaeology. The project is a collaboration between the UNESCO Center for the Researching and Conserving of Vietnamese Antiquities, Hanoi University of Mining and Geology, the Vietnamese Institute of Archaeology, and Nam Can Tho University.

Keywords— Ancient glass classification, non-destructive scanning electron microscopy (SEM), machine learning in archaeology